

Working with Altera Quartus

Creating a new Project

1. From Desktop / search for quartus-> Open Quartus.(8.1) [**DO NOT OPEN Quartus 11**]
2. Click File->New Project Wizard->Next
3. Fill out the following:
 - Working directory? – Browse->Create a folder in a directory **other than C drive** (**Downloads, Desktop are parts of C drive, DO NOT OPEN A PROJECT THERE!**).
 - *The name of the folder should **not** start with a number or any special character!-> Select the folder*
 - What is the name of the **project**->Type a name and remember it (Say “expt1”). Same name is going to be copied to the next box automatically.[**DO NOT start the name with any special character or number**]
 - Press next.
4. Press next **again**
5. Select Device Family: **FLEX10KE** and press next.
6. Fill out the following in **all three pairs of boxes**:
Tool name: 1. Custom / 2. Custom / 3. Custom
Format: 1. Verilog HDL/ 2. Verilog/ 3. Verilog
Press Next->Finish

Creating a New Verilog file:

7. From the Quartus application->File->New->Verilog HDL File->OK
8. Write the code and save it with the same name as that given in expt1 with extension of .v (Example: expt1.v)
9. The **project name** [expt1] and the **module name** [expt1] should be the same.
10. **COMPILE** the code.

Viewing RTL Circuit

** RTL circuit: **Tools-> Netlist viewer->RTL viewer.**

Output: Creating a new vector Waveform File & Simulation

11. File->New->Vector Waveform File
12. Save as-> Save with the **same filename** as the .vwf file (Example: expt1.vwf)
13. Right click on Name->Insert->Insert Node or BUS
14. Click Node Finder (**Make sure** that you’ve compiled the code earlier in step 10).
Filter: Pins: all
Look in: Filename (Example expt1.v)
Click list->Click “>>” -> OK -> OK
15. Right-click on each **input**->Value->Clock and set up the clocks.
16. From the menu -> Assignment->Settings->Simulator Settings
 - Simulation Mode: **Functional**
 - Click OK.
17. From the menu -> Processing-> Generate Functional Simulation Netlist
18. Processing-> Start Simulation

Some Extra information

19. For inserting custom inputs in the vector waveform file, watch this tutorial :

<https://youtu.be/54F7N2vncLY?feature=shared>

20. For changing grid size: go to Edit > Grid Size > change the period of each grid

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